

9 (a) The passage outlines the endocrine system.

Complete the passage by using the most appropriate scientific terms.

The endocrine system consists of tissues and , which secrete hormones such as insulin, glucagon and antidiuretic hormone (ADH). Glucagon only affects target cells that have complementary receptors, which are located on the Glucagon binds to these receptors and this leads to the production of a , which transfers the signal throughout the cytoplasm. The target cells for insulin are in the liver, and adipose tissue. The target cells for ADH are those of the distal convoluted tubule and the of the kidney nephron. These hormones are involved in , which is the maintenance of a constant internal environment. [6]

(b) The blood glucose concentration of a person was measured at regular intervals after the ingestion of a meal rich in glucose.

Fig. 9.1 shows the results of this investigation.

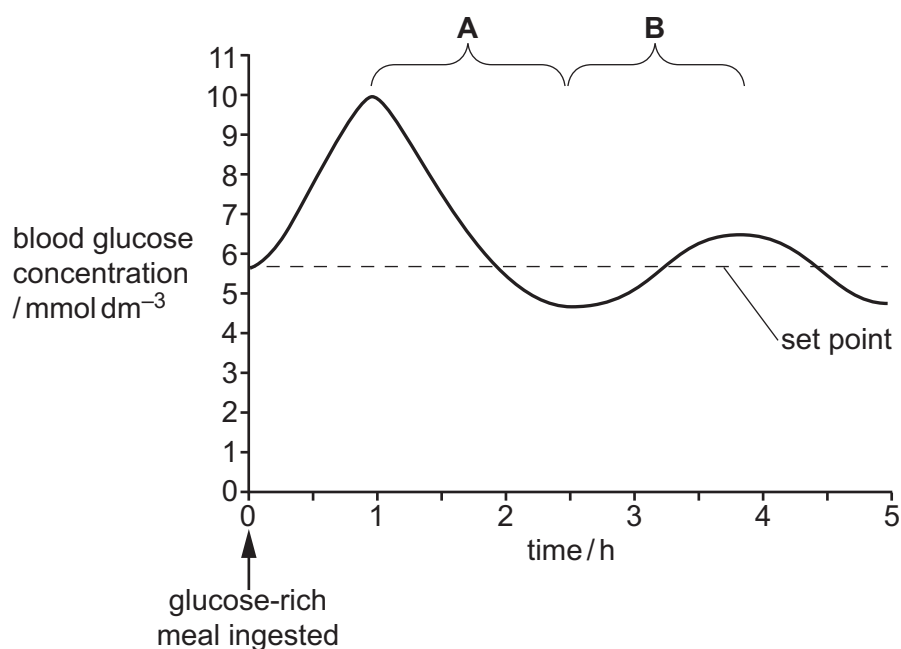


Fig. 9.1

- (i) Explain how the blood glucose concentration is reduced during phase **A** of the curve.

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..... [3]

- (ii) Suggest why the blood glucose concentration increases again during phase **B**.

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..... [1]

[Total: 10]